

Modeling Enteric and Dung Methane Emissions in Yak Grazing Systems Under Shared Socioeconomic Pathways: A Literature-Grounded Agent-Based Approach with Applications for the Kyrgyz Republic

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Abstract

Enteric fermentation and manure management are major sources of anthropogenic methane, yet most Tier 2-style livestock emissions models are parameterized for cattle rather than yak (*Bos grunniens*), a species with documented differences in rumen microbiology, energy metabolism, and pastoral management. We present a NetLogo agent-based model adapting a generic cattle enteric-methane framework to reflect yak-specific physiology, demography, and grazing-system emission sources, using published measurements primarily from the Qinghai-Tibetan Plateau (QTP). The model implements a standard IPCC/NRC Tier 2 energy-balance approach for enteric methane, a newly added dung-deposition flux module, and a density-dependent demographic submodel, coupled to an illustrative simplified climate response chain. We document the equations used, the literature basis for each parameter, structural bugs identified during implementation and their corrections, and a validation check against an independent field-measured benchmark. We conclude by discussing the model’s status as a proof-of-concept and outlining what would be required to recontextualize it for the Kyrgyz Republic, a Central Asian yak-herding country with distinct geography, herd scale, and management institutions from the QTP source literature.

1 Introduction

Livestock enteric fermentation is estimated to contribute a substantial share of anthropogenic methane emissions [IPCC, 2019]. Yak, a domesticated bovine adapted to high-altitude, cold, hypoxic environments, is understudied relative to cattle in national greenhouse gas inventory frameworks, despite evidence that its digestion and metabolism differ meaningfully from *Bos taurus* [Ding et al., 2010]. Applying cattle-calibrated Tier 2 parameters to yak populations risks systematic misestimation of emissions.

This report documents the development of an agent-based simulation model, built in NetLogo, that adapts a pre-existing cattle enteric-methane toy model to a yak-specific parameterization. The model integrates (i) a Tier 2 energy-balance enteric methane calculation, (ii) a newly introduced dung/manure methane flux module, (iii) a demographic submodel governing herd growth and mortality, and (iv) a simplified illustrative climate response chain linking cumulative emissions to atmospheric methane concentration and temperature anomaly under Shared Socioeconomic Pathway (SSP) scenarios.

2 Methods

This section documents every equation implemented in the model, organized by subsystem. Equations are given in their implemented form, variables are defined, and the literature basis is cited where one exists. Equations without a species-specific literature basis are explicitly marked as structural or illustrative rather than empirically derived.

2.1 Enteric Methane Emission: Tier 2 Energy-Balance Framework

The general Tier 2 / NRC net-energy structure is not itself yak-specific; it is the standard framework applied across ruminant species [IPCC, 2019, Global Research Alliance, 2020]. Yak-specific literature was used to adjust the *parameter values* populating this structure (methane conversion factor, digestible energy, maintenance energy) rather than the structural form of the equations.

2.1.1 Net energy for activity

$$NE_a = C_a \times NE_{ma} \quad (1)$$

where C_a is a coefficient reflecting feeding/grazing situation (grazing, supplemented-grazing, indoor) and NE_{ma} is net energy for maintenance. Feeding-situation categories were set to mirror the experimental design of Ding et al. [2010], who compared grazing, indoor oat-hay, and indoor concentrate-supplemented diets.

2.1.2 Net energy for growth

$$NE_G = 22.02 \times \left(\frac{BW}{0.8 \times BW_{mature}} \right)^{0.75} \times G^{1.097} \quad (2)$$

where BW is current live body weight (kg), BW_{mature} is mature/reference body weight (kg), and G is average daily weight gain (kg/day). This is the standard NRC growth-energy equation. **Implementation note:** an earlier version of the code omitted the exponent 0.75 and substituted a body-condition score for BW_{mature} , producing non-physical gross energy values (validation ratio ≈ 38 , Section 4); the corrected form above restores the proper exponent and introduces a distinct mature-weight parameter.

2.1.3 Net energy for lactation

$$NE_L = \left(\frac{Milk_{annual}}{365} \right) \times (1.47 + 0.40 \times Fat\%) \quad (3)$$

Milk-yield inputs are cross-checked against yak-specific lactation data: average milk yields of 0.7 ± 0.2 kg/day over a 109 ± 9 day milking period under once-daily milking, versus 1.24 ± 0.3 kg/day over 127 ± 6 days under twice-daily milking [Long et al., 1999].

2.1.4 Net energy for work and pregnancy

$$NE_W = 0.10 \times NE_{ma} \times H, \quad NE_P = 0.10 \times NE_{ma} \quad (4)$$

where H is hours worked per day. Both coefficients are generic Tier 2 constants, not yak-specific, retained for draft/pack-yak use cases.

2.1.5 Efficiency ratios for maintenance and growth

$$REM = 1.123 - (4.092 \times 10^{-3} \times DE) + (1.126 \times 10^{-5} \times DE^2) - \frac{25.4}{DE} \quad (5)$$

$$REG = 1.164 - (5.160 \times 10^{-3} \times DE) + (1.308 \times 10^{-5} \times DE^2) - \frac{37.4}{DE} \quad (6)$$

where DE is dietary digestible energy (% of gross energy). These are standard IPCC/NRC coefficient equations. Digestible energy values used (65/55/45% for high/moderate/low diet quality) were set lower than typical feedlot-cattle defaults to reflect grazing alpine-meadow forage, consistent with the QTP grazing/indoor cold-season study design of Ding et al. [2010], though the specific constants are reasoned approximations pending direct local measurement.

2.1.6 Gross energy intake

$$GE = \frac{\left(\frac{NE_{ma} + NE_a + NE_L + NE_W + NE_P}{REM} \right) + \left(\frac{NE_G}{REG} \right)}{DE/100} \quad (7)$$

2.1.7 Methane emission factor

$$EF = \frac{GE \times (Y_m/100) \times 365}{55.65} \quad (8)$$

where Y_m is the methane conversion factor (Section 2.2) and 55.65 MJ/kg is the energy content of methane [AgLEDx, 2020]. This yields annualized methane production (kg CH₄/year).

2.1.8 CO₂-equivalent conversion

$$EF_{CO2eq} = EF \times 28 \quad (9)$$

using a 100-year GWP of 28 for methane (AR5), applied uniformly and not species-specific.

2.2 Methane Conversion Factor (Y_m)

$$Y_m = Y_{m,base} - \delta_{supplement} + \epsilon \quad (10)$$

$Y_{m,base} = 5.4\%$ is drawn directly from Ding et al. [2010], who measured an average proportion of gross energy lost as CH₄ of 54 ± 15.2 kJ/MJ under QTP grazing and indoor cold-season conditions (equivalent to $5.4\% \pm 1.5\%$ of GE intake), and found that CH₄ production declined with forage:concentrate ratios of 0.6:0.4 and 0.4:0.6 relative to an oat-hay-only treatment. This value is markedly below the cattle IPCC default of 6.5%, consistent with mechanistic evidence of lower H₂ production in yak rumen fluid relative to cattle [Mi et al., 2017] and enrichment of methanogenesis pathways in the cattle metagenome relative to yak [Zhang et al., 2016].

$\delta_{supplement} = 0.8$ is applied under supplemented-grazing conditions, reflecting the qualitative finding that supplemental concentrate feeding markedly lowered enteric methane emissions in grazing yaks during the warm season [Ma et al., 2025]; the magnitude is a reasoned approximation, not a directly cited numeric offset.

$\epsilon \sim \mathcal{N}(0, 1.5) \times 0.1$ introduces stochastic variability matching the reported standard deviation in Ding et al. [2010].

2.2.1 Validation benchmark

$$\text{validation-ratio} = \frac{EF_{CO_2eq}/28}{28} \quad (11)$$

comparing modeled annual per-animal CH_4 output (kg/year) against the field benchmark of approximately 28 kg CH_4 /year for a 175 kg grazing non-lactating yak, measured via the SF_6 tracer gas technique [Ding et al., 2010]. An independent cross-check reported elsewhere in the literature — yak methane emissions measured at 39% lower than IPCC Tier 2 model predictions [Wang et al., 2025] — is noted as an external reference but not built into the code as an active constraint.

2.3 Dung/Manure Methane Flux

$$Flux_{dung}(t) = \phi \times e^{-0.3t} \quad (12)$$

where ϕ is a calibration constant (**dung-CH4-per-patch**) and t is ticks since deposition. This exponential-decay structure reflects the qualitative finding that dung deposition significantly increases CH_4 emissions due to dissolved CH_4 , large microbial populations, highly degradable organic compounds, and anaerobic conditions in fresh patches, with flux driven by moisture content and surface temperature [Liu et al., 2017]. The decay rate and magnitude constant are calibration targets rather than directly cited values. The calibration reference target is that CH_4 and N_2O contributed 21.1% and 44.8% respectively to a total annual global warming potential (GWP) budget of 334.2 tonnes CO_2 -eq from family-farm grazing yaks [Liu et al., 2017].

2.4 Demographic Submodel

2.4.1 Density-dependent reproduction

$$P(\text{calve}) = r \times \max\left(0, 1 - \frac{N}{K}\right) \quad (13)$$

where r is the base reproductive rate, N is current population count, and K is carrying capacity (a multiple of patch count). This density-dependence term is a generic ecological mechanism added to correct unconstrained exponential growth observed in initial testing (population reaching $\sim 92,000$ from a starting value of 100 over 85 simulated years).

The base reproductive rate ($r = 0.44$) reflects a regional statistic from Hongyuan County, Sichuan (1976–1980), in which 43.8% of nearly half a million breeding-age female yak produced a calf that also survived in a given year [Food and Agriculture Organization, 2003]. Minimum age at first reproduction (3 years) reflects that yaks are usually first mated at 3 to 4 years of age [IVIS, 2002], and the minimum one-year inter-calving interval reflects that female yak most often calve once every two years, or twice in three years, producing on average four to five calves in a lifetime [Food and Agriculture Organization, 2003].

2.4.2 Mortality

$$P(\text{death}) = \begin{cases} m_{calf} & \text{if age} < 1 \\ m_{adult} & \text{otherwise} \end{cases} \quad (14)$$

m_{calf} (default 0.15) is bracketed by calf survival rates of 94% and 81% observed across two successive generations in an inbreeding study, implying first-year mortality of roughly 6–19% [Food and Agriculture Organization, 2003]. m_{adult} (default 0.05) has **no yak-specific literature source** identified in this review and is retained as a free calibration parameter.

2.5 Climate Response Chain (Illustrative)

The following equations govern a simplified atmospheric methane concentration and temperature response. **None of these equations are derived from yak or livestock literature**; they constitute a generic, illustrative reduced-form climate box model included to visualize downstream effects, and should not be presented as a validated climate model.

$$\dot{C} = \beta \times RF + B \times EF_{cumulative} - \sigma \times C \quad (15)$$

$$C_{t+1} = C_t + \dot{C} \quad (16)$$

where C is atmospheric methane concentration (ppb, initialized at pre-industrial baseline $C_{pi} = 722$ ppb), RF is the selected SSP radiative forcing level, and β, σ, B are scenario-dependent coefficients.

$$\dot{T} = \mu \times \ln\left(\frac{C}{C_{pi}}\right) - \alpha \times T \quad (17)$$

$$T_{t+1} = T_t + \dot{T} \quad (18)$$

a logarithmic forcing-response relationship loosely analogous to standard radiative forcing formulations, with illustrative (not empirically fitted) coefficients μ, α varying by SSP scenario (SSP1-1.9 through SSP5-8.5, forcing magnitudes 1.9–8.5 W/m²).

3 Model Implementation: Bugs Identified and Corrections

Table 1 summarizes structural and mechanical errors identified during implementation and testing, distinct from the parameter-value adaptations described in Section 2.

None of these were errors in the literature-derived parameter values themselves; Y_m and $DE_{percentage}$ returned expected magnitudes throughout. The errors were confined to variable declaration, wid-get/code name-matching, and one un-exponentiated growth-energy term inherited from the original cattle-model code.

4 Results: Interface Parameters and Validation

Table 2 lists all interface-exposed variables, their ranges, defaults, units, and the rationale/evidentiary basis for each.

Variable	Default	Units	Basis / purpose
number-of-yaks	100	head	Starting herd size
reproductive-rate	0.44	prob./yr	Hongyuan County statistic [Food and Agriculture Organization, 2003]
mortality-rate-calf	0.15	prob./yr	Bracketed by 81–94% calf survival [Food and Agriculture Organization, 2003]
mortality-rate-adult	0.05	prob./yr	Uncalibrated – no citation found
avg_live_body_weight	475	kg	Matches growing yak steers in Ding et al. [2010]

Variable	Default	Units	Basis / purpose
mature-weight	280	kg	Placeholder estimate, not directly cited
avg_daily_weight_gain	0.1	kg/day	Range allows negative values for winter weight loss
Y_m (monitor)	–	% of GE	Benchmarked to Ding et al. [2010] ($5.4 \pm 1.5\%$)
dung-CH ₄ -per-patch	0.5	kg CH ₄ /patch/tick	Calibration constant, target ratio from Liu et al. [2017]
feeding-situation	grazing	categorical	grazing / supplemented-grazing / indoor-oat-hay
SSP-Scenario	SSP_2-4.5	categorical	Standard IPCC SSP forcing scenarios
validation-ratio (monitor)	–	ratio	Target ≈ 1.0 against Ding et al. [2010] benchmark

Table 2: Selected interface variables (abbreviated; full slider/chooser/monitor list omitted for brevity).

5 Discussion: Limitations

Several parameters carry no direct literature support and are flagged rather than presented as evidence-based:

- **mortality-rate-adult** and **mature-weight** require independent calibration, e.g. against total Plateau population estimates of approximately 16 million yaks in China [Zhu et al., 2024], over 90% of the global yak population, or approximately 13 million yaks on the QTP specifically [Zhou et al., 2017].
- The climate response chain (Section 2.6) is illustrative only and not derived from yak or livestock literature.
- The dung-flux magnitude constant is a calibration target; Liu et al. [2017] report aggregate system-level fluxes, not a universal per-patch constant transferable across stocking densities.

6 Recontextualization for the Kyrgyz Republic

The literature underlying every yak-specific parameter in this model — Y_m , $DE_{percentage}$, dung-flux ratios, reproductive rate — originates from Chinese Qinghai-Tibetan Plateau research stations. None was measured on Kyrgyz herds, pastures, or under Kyrgyz management systems. This model demonstrates that a yak-specific enteric-and-dung emissions and demography model is technically feasible and produces plausible, literature-anchored magnitudes; it does not demonstrate that the specific parameter values transfer to Central Asia.

6.1 Scale and Recency

Yak husbandry in Kyrgyzstan has a roughly 100-year history and has grown, with government subsidy support, to a total population of approximately 100,000 animals managed by local pasture

Table 1: Implementation bugs and corrections

Bug		Cause	Fix
<code>emission-rate</code> ready defined"	"al-	Turtle-owned variable name collided (case-insensitively) with the <code>Emission-Rate</code> interface slider	Renamed turtle variable to <code>enteric-emission-rate</code>
<code>food-available</code> unde-fined	unde-	Used as a monitor but never declared in <code>globals</code>	Declared in <code>globals</code> ; initialized in <code>setup</code>
Missing plot pens		Code referenced pens " <code>enteric</code> "/" <code>dung</code> " not present in the plot widget	Pens added manually in the plot editor; existence confirmed via <code>plot-pen-exists?</code>
<code>validation-ratio</code> = 0		Population had collapsed or an input chain was broken	Diagnosed via hardcoded known inputs in <code>setup</code> with intermediate variable printing
Unconstrained population growth (100 → 92,247 over 85 years)		No density-dependent limit on reproduction	Added carrying-capacity term (Eq. 13)
<code>validation-ratio</code> ≈ 38	≈	NE_G term missing the 0.75 exponent; wrong reference variable used in place of mature body weight	Corrected exponent and introduced distinct <code>mature-weight</code> parameter (Eq. 2)

committees [NABU] — roughly two orders of magnitude smaller than the QTP source system (13–16 million head).

6.2 Geography and Seasonality

The Kyrgyz Republic comprises the western Tien-Shan and northern Pamir/Alay ranges, with over 90% mountainous terrain and most valleys at 600–1,400 m elevation [Kudaliauskaite et al., 2019], with peaks exceeding 7,000 m [Kreutzmann, 2020] — a distinct massif from the QTP with different precipitation and vegetation regimes. Notably, winter fodder is comparatively less limited for Kyrgyz herders in the eastern Pamir due to lower snow accumulation despite extreme temperatures, permitting winter grazing at higher altitudes without supplementary fodder [Kreutzmann, 2020] — a materially different seasonal constraint than the QTP’s deep-snow winter scarcity regime that calibrated the model’s current seasonal forage function.

6.3 Herd Composition

Kyrgyz yaks are typically grazed within mixed-species herds; one local pasture inventory recorded 6,517 yaks among 20,452 total livestock [CAMP Alatau], and Kyrgyzstan’s pastoral system broadly relies on livestock diversity across cattle, horses, sheep, and goats sharing pasture resources [IUCN, 2020]. The current model’s single-species forage-competition assumption would require a multi-species extension to be representative.

6.4 Governance

Kyrgyz yak populations are managed by local pasture committees [NABU], with survey evidence indicating that over 90% of respondents report increasing local livestock numbers year-on-year, attributed roughly evenly to subsistence needs and income generation [CAMP Alattoo] — a commons-governance dynamic distinct from the household/family-farm framing underlying the QTP dung-emissions source study [Liu et al., 2017].

6.5 Transferability Assessment

The structural form of the Tier 2 energy-balance equations, the dung-decay mechanism, and the density-dependent demographic logic are reusable across contexts. Specific numeric constants — Y_m , the seasonal forage curve, reproductive/mortality rates, and dung-flux magnitude — require re-derivation from Kyrgyz-specific data. One plausible point of cross-context comparability is stocking density: a local norm of one conditional head per hectare used in Kyrgyz pasture load assessments [CAMP Alattoo] is numerically similar to the QTP’s reported sustainable stocking rate of approximately one yak per hectare, though this warrants direct testing rather than assumption.

7 Conclusion

This model constitutes a proof-of-concept demonstrating that an agent-based, literature-grounded methane and demography model is achievable for yak systems using currently published parameters, which are concentrated in Chinese QTP research given the relative scarcity of yak methane studies elsewhere. Recontextualizing the model for the Kyrgyz Republic, or any other yak-herding region, is a distinct exercise requiring locally sourced measurements of methane conversion factors, seasonal forage dynamics, demographic rates, and stocking practices, rather than a direct parameter transfer.

8 Preliminary Results: Feeding-System Mitigation Across SSP Scenarios

To illustrate the model’s intended use case, a BehaviorSpace experiment was run crossing two feeding-system configurations (**grazing**, **supplemented-grazing**) against four SSP radiative-forcing scenarios (SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5), with 10 stochastic repetitions per combination over an 85-year horizon (2015–2100). Given the known calibration issues described in Sections 3 and 5, results are interpreted qualitatively (direction and relative magnitude) rather than as validated absolute emission estimates.

8.1 Mitigation direction is consistent but modest in current calibration

Figure ?? shows cumulative enteric $\text{CH}_4\text{-CO}_2\text{eq}$ under grazing versus supplemented-grazing, averaged across all four SSP scenarios. The supplemented-grazing trajectory tracks consistently below the grazing baseline across the full simulation horizon, consistent with the direction reported in Ding et al. [2010] and Ma et al. [2025]. However, the separation between the two curves is small relative to the total magnitude ($\sim 3\text{--}5\%$ by year 85), smaller than the qualitative reductions reported in the source literature. This is consistent with the model’s current mis-calibration (Section 4): the $\delta_{\text{supplement}} = 0.8$ percentage-point offset to Y_m is a fixed, reasoned approximation rather than a fitted value, and its effect is proportionally diluted by the uncorrected NE_G term inflating the denominator of the emissions calculation.

8.2 SSP scenario has negligible effect on herd-level emissions

Figure ?? disaggregates the feeding-system comparison by SSP scenario. The mitigation gap between feeding systems is visually indistinguishable across all four SSP pathways. This is an expected structural property of the current model rather than a finding: SSP scenario enters only through the climate response chain (Eqs. 15–18), which is downstream of and does not feed back into the herd demographic or enteric-emission submodels. The near-identical temperature trajectories across all four SSP scenarios shown in Figure ?? confirm this — cumulative herd-level $\text{CH}_4\text{-CO}_2\text{eq}$ output (driven by unconstrained population growth; see Section ??) dominates the radiative-forcing term in Eq. 15, effectively saturating the climate submodel’s sensitivity to the SSP coefficient set $(\beta, \sigma, \mu, \alpha)$ at this population scale. This indicates that meaningful SSP-differentiated climate output would require either the demographic carrying-capacity correction described in Table 1 (to prevent population, and hence $EF_{\text{cumulative}}$, from dominating Eq. 15) or a rescaling of the B coefficient.

8.3 Validation ratio confirms a systematic, not drifting, offset

Figure ?? shows the validation ratio holds essentially flat at $\approx 37\text{--}38$ across the full 85-year horizon for both feeding systems, with narrow variance bands. The flatness of this ratio over time is informative: it indicates the mis-calibration identified in Section ?? (the un-exponentiated NE_G term) acts as a consistent multiplicative offset rather than a compounding or time-dependent error. This supports treating *relative* comparisons between scenarios (Sections 8.1–8.2) as more robust than the model’s current *absolute* emission magnitudes, pending the correction described in Table 1.

9 Outlook: Applicability of this Model Structure to the Kyrgyz Republic

The preliminary results above are offered as a demonstration of model *behavior*, not as Kyrgyz-relevant estimates — none of the parameters populating this run are Kyrgyz-derived (Section 7). With that caveat, three structural features of the current implementation are directly relevant to a future Kyrgyz-parameterized version:

- **The mitigation-comparison structure is directly reusable.** The feeding-system chooser mechanism (grazing / supplemented-grazing / indoor-oat-hay) that drove the comparison in Figure ?? maps naturally onto a Kyrgyz policy question of practical interest: whether pasture-committee-level decisions about supplementary feeding during winter grazing (Section 7.2) could measurably reduce herd-level methane output. Re-running this same experiment design with Kyrgyz-calibrated Y_m , $DE_{\text{percentage}}$, and $\delta_{\text{supplement}}$ values, once available, would produce a policy-legible mitigation estimate at the scale of a single pasture committee’s herd.
- **The demographic submodel requires the carrying-capacity fix before Kyrgyz-scale runs are meaningful.** At $\sim 100,000$ head nationally versus 13–16 million on the QTP (Section 7.1), Kyrgyzstan’s much smaller population makes the model considerably more sensitive to getting density-dependent reproduction right; the unconstrained-growth bug would distort a Kyrgyz-scale run more severely, proportionally, than it does the current QTP-scale test run.
- **The SSP/climate module is of limited direct relevance at Kyrgyz herd scale.** Given the findings that SSP scenario has negligible effect on herd-level emissions in the

current model, and given that Kyrgyzstan’s ~100,000-head national herd is roughly two orders of magnitude smaller than the QTP system this climate chain was tested against, the illustrative climate submodel is unlikely to be a productive extension target for a Kyrgyz-focused version of this tool. Effort would be better directed at the enteric and dung-emission submodels (which are directly policy-actionable at pasture-committee scale) and the multi-species forage-competition extension noted in Section 7.3, rather than at refining the climate response chain.

In summary, this proof-of-concept demonstrates that the model’s *comparative* mitigation-scenario structure is sound and portable; the priority for a Kyrgyz adaptation is (i) correcting the two implementation bugs noted in Table 1, (ii) sourcing Kyrgyz-specific Y_m , demographic, and seasonal-forage parameters, and (iii) extending the single-species assumption to reflect Kyrgyzstan’s mixed-species grazing system (Section 7.3), rather than further development of the illustrative climate-response chain.

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